MILITARY SPECIFICATION

ARMAMENT SUBSYSTEM, HELICOPTER: M28A1

This specification is approved for use by Rock Island Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of helicopter armament subsystem consisting of a flexible, power-operated, remotely-controlled turret with control circuits to permit operation of the subsystem in response to an optical sighting unit and forward firing utilizing the pilot/gunner's cyclic stick. The helicopter armament subsystem is hereinafter referred to as the subsystem.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-45924	-	Control Panel, Gunner
MIL-E-45925	-	Electronic Components Assembly, (11688148)
MIL-D-45926	-	Drum, Ammunition, Rotary 40mm
MIL-S-45927	-	Sighting Station
MIL-D-45929	-	Drum Ammunition, Rotary, 7.62mm

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MIL-C-45953	-	Control Panel, Pilot
MIL-C-45954	-	Controller, Weapons Gun Speed and Launcher Brake, RH
MIL-C-45956	-	Controller, Weapons Gun Speed and Launcher Brake, LH

STANDARDS

Federal		
Fed. Std. No. 356	-	Commercial Packaging of Supplies and Equipment
Military		
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-721	-	Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors and Safety
MIL-STD-781	-	Reliability Tests, Exponential, Distribution
MIL-STD-810	-	Environmental Test Methods

DRAWINGS

Rock Island Arsenal		
F11690269	-	Armament Subsystem, Helicopter, 7.62mm High Rate Gun/40MM Grenade Launcher, M28A1

(Product drawings referenced in this specification form a part of F11690269.)

PUBLICATIONS

Rock Island Arsenal		
P11691612-1	-	Packaging Data Sheet for Armament Subsystem, Helicopter, M28A1 W/E (Config. B)
QE-G-DL 11690269	-	Quality Engineering Data List-Armament Subsystem, Helicopter, 7.62mm High Rate Gun/40MM Grenade Launcher, M28A1

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified (see 6.2). Unless otherwise specified (see 6.2), the first article shall include a pilot pack (see 5.1).

3.2 Material, construction and design. Subsystems and parts shall conform to the materials, construction and design requirements specified herein, on Drawing F11690269, and MIL-W-13855.

3.2.1 Ammunition drums.

3.2.1.1 The 40mm ammunition drum shall be in accordance with MIL-D-45926.

3.2.1.2 The 7.62mm ammunition drum shall be in accordance with MIL-D-45929.

3.2.2 Gunner control panel. The gunner control panel shall be in accordance with MIL-C-45924.

3.2.3 Pilot control panel. The pilot control panel shall be in accordance with MIL-C-45953.

3.2.4 Weapon controllers. The weapon controllers shall be in accordance with MIL-C-45954 and MIL-C-45956.

3.2.5 Sighting station. The sighting station shall be in accordance with MIL-S-45927.

3.2.6 Electronic components assembly (ECA). The electronic components assembly shall be in accordance with MIL-E-45925.

3.2.7 Weight. The overall weight of the complete subsystem, including all Government-furnished equipment (GFE) and a full complement of ammunition (4,000 rounds per drum of 7.62mm ammunition and 250 rounds per drum of 40mm ammunition) shall not exceed the following:

Twin 40mm weapon configuration	885 pounds
One 7.62mm and one 40mm weapon configuration	930 pounds
Twin 7.62mm configuration	975 pounds

3.3 Functional requirements.

3.3.1 Electrical power. The subsystem shall operate from the power sources listed in 3.3.1.1, 3.3.1.2 and 3.3.1.3.

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3.3.1.1 28 (22 to 30) volts direct current (vdc) power source. With the subsystem operating in any of the following configurations, the maximum current consumption shall not exceed the value listed.

Mixed 40mm and 7.62mm	20 amperes
Dual 40mm	10 amperes
Dual 7.62mm	35 amperes

3.3.1.2 24 vdc battery power source. With the subsystem operated in any of the following configurations, the average current consumption (excluding initial starting current surge) during steady-state firing shall not exceed the value listed.

Configuration	Steady State Average Current Consumption
Mixed:	
40mm	90 amperes
7.62mm Low Rate	50 amperes
7.62mm High Rate	100 amperes
Dual:	
40mm	180 amperes
7.62mm	200 amperes

NOTE: A required input voltage of 25 ± 0.5 vdc at the weapons controller input (see 3.3.16) is specified for test purposes only to check weapons' rate of firing and average steady-state current consumption specified above. It is not to be construed as placing a tolerance on the aircraft battery. The aircraft battery is not a regulated device and its voltage level is determined by the applied load, state of charge, etc.

3.3.1.3 115 volt alternating current (vac) power source. With 115 + 4.5 - 7.5 vac at 400 hertz (Hz) ± 20 Hz applied to the subsystem the maximum root-mean-square (rms) current consumption shall not exceed one ampere.

3.3.2 Hydraulic power. The turret, powered by a hydraulic motor in azimuth and a hydraulic actuator in elevation, shall perform as specified herein when supplied with a hydraulic pressure of 1500 ± 25 pounds per square inch gage (psig). Flow rate, as supplied by a constant pressure variable displacement hydraulic pump, shall not exceed 5.5 gallons per minute (gpm) at the specified pressure. Hydraulic fluid shall be in accordance with MIL-H-5606.

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3.3.2.1 Hydraulic proof pressure. The subsystem shall withstand a proof pressure of 2225 ± 25 psig applied to the pressure lines with a 200 ± 20 psig back pressure applied to the return lines for a period of 2 minutes before and after operating the subsystem throughout its full azimuth, elevation and depression travel. There shall be no evidence of external leakage other than a slight wetting insufficient to form a drop.

3.3.3 Subsystem controls and indicators.

3.3.3.1 Pilot's control panel. The controls and indicators on the pilot's control panel shall provide the following functions:

a. Pilot/gunner switch. The pilot/gunner switch in the pilot position shall render the gunner's cyclic switch and sighting station inoperative and the weapons shall be operated by the pilot's cyclic stick in the fixed forward position only. In the gunner position, the triggers on the pilot's and gunner's cyclic sticks shall be inoperative and the subsystem shall be controlled by the gunner's sighting station.
b. Master arm switch. The master arm switch in the safe position shall energize the turret electrical system and hydraulic system, and the turret shall drive into a stow position. The turret can then be operated in a flexible mode through use of the sighting station by the gunner when selected by the pilot. The weapons shall not operate by trigger actuation until the master switch is placed in the armed position, which shall electrically energize the weapons and permit operation by the gunner or pilot depending upon the position of the pilot/gunner switch.
c. Weapon select switch. The pilot's WEAPON SELECT switch shall select either the right hand or left-hand weapon for firing, when actuated to either the RIGHT or LEFT position. When actuated to the BOTH position with similar weapons installed on the left and right-hand position, both weapons shall fire simultaneously upon trigger actuation. When actuated to the BOTH position, with a machine gun and grenade launcher installed, the grenade launcher firing circuit shall be inoperative and the machine gun alone shall fire upon actuation of the pilot's cyclic stick trigger.
d. System armed indicator. The system armed indicator (amber) shall be illuminated when the subsystem is armed and ready for weapon operation.
e. Safe indicator. The safe indicator (green) shall be illuminated when the armament turret has been electrically energized, but not armed.
f. Weapon indicators. The weapon indicators (blue) shall be illuminated to indicate the type and position of the weapons installed in the turret.
g. Point/area fire. The point/area fire switch shall produce machine gun oscillation as defined in 3.3.11.

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3.3.3.2 Gunner's control panel. The controls and indicators on the gunner's control panel shall provide the following functions:

a. Clear/unclear switch. The clear/unclear switch in the clear position shall cause the machine gun to be automatically cleared at the end of each firing. In the unclear position, the machine gun shall not be cleared, and live rounds shall remain in the weapon upon cessation of firing.
b. Fire out switch. The fire out switch in the ON position shall permit the last rounds of ammunition contained in the flexible chute to be fired. In the OFF position, the operation of each weapon shall terminate automatically when the last round passes the LAST ROUND switch located in the appropriate ammunition feed unit (either the 40mm drum or the 7.62mm drum).
c. Wing stores select switch. The wing stores select switch shall permit the gunner to energize circuitry to select either the inboard or outboard wing weapons.
d. Pilot override switch. The pilot override switch shall permit the gunner to override the pilot's controls (see 3.3.3.7).
e. Rounds counters. Rounds counters shall indicate the percentage of rounds remaining in the ammunition drum.
f. Unclear weapon indicator. The unclear weapon indicator (amber) shall be illuminated when the machine gun is not clear.
g. System armed indicator. The system armed indicator (amber) shall be illuminated when the subsystem is armed and ready for weapon operation.
h. Safe indicator. The safe indicator (green) shall be illuminated when the armament turret has been electrically energized, but not armed.
i. Gunner control indicator. The gunner control indicator (blue) shall be illuminated when the subsystem is placed in the gunner's control or when the pilot override switch is actuated.
j. Weapon indication. The weapon indicator (blue) shall be illuminated to indicate the type and position of the weapons installed in the turret.
k. Point/area fire. The point/area fire switch shall produce machine gun oscillation as defined in 3.3.11.

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3.3.3.3 Gunner's sighting station. The controls and indicators on the gunner's sighting station shall provide the following functions:

a. Weapon select switch. A gunner's WEAPON SELECT switch shall actuate the right-hand weapon in the RIGHT position and the left-hand weapon in the LEFT position. When actuated to the both position, with similar weapons installed in the left and right-hand positions, both weapons shall fire simultaneously upon trigger actuation. When actuated to the BOTH position with a machine gun and grenade launcher installed, the first trigger actuated shall fire the respective weapon which shall continue firing until the trigger switch is released regardless of any other concurrent trigger switch action (see "c" below).
b. Sight action switches. Actuation of either the left or right-hand switch shall cause the turret to move in azimuth and elevation in response to positioning of the sighting station controller.
c. Trigger switches. With the left or right-hand sight action switch depressed, the first position of either of the index finger trigger switches shall result in low speed operation of the 7.62mm gun or guns. Depressing either of the index finger trigger switches into the second position will result in the 7.62mm gun or guns to operate at maximum speed. Depressing either of the thumb switches with either sight action switch depressed, shall result in 40mm weapon operation.
d. Compensation switch. The compensation switch shall permit the gunner the choice of utilizing or not utilizing the velocity and range compensation.
e. Range potentiometer. The potentiometer shall permit the gunner to add the estimated range compensation when the compensation switch is activated.
f. Reticle switch. The reticle switch shall select either of the reticle illumination lamp filaments.
g. Reticle intensity rheostat. The adjustable rheostat shall control intensity of the reticle illumination lamp from the off position to full brightness.
h. Pilot control indicator. The pilot control indicator (amber) shall be illuminated when the subsystem is in the pilot control mode.

3.3.3.4 Cyclic stick control. The subsystem shall fire from either the gunner's cyclic stick or the pilot's cyclic stick. These two position trigger switches shall control low speed machine gun operation when depressed to the first position and maximum speed when depressed to the second position at the rated speeds (see 3.3.16.a). The grenade launcher shall operate at rated speed (see 3.3.16.b) in either position of the cyclic stick trigger.

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3.3.3.5 Pilot range compensation. The subsystem shall have range compensation through the use of the pilot controlled potentiometer. The range compensation shall be demonstrated by simulating the variable potentiometer. The range potentiometer shall adjust the turret elevation fixed forward position to compensate for variation in target range according to 3.3.6.1.

3.3.3.6 Velocity compensation. The subsystem shall provide azimuth compensation to the turret based on an automatic input from the aircraft pitot tube. This capability shall be demonstrated by applying voltage corresponding to the transducer output at the transducer connector on the electronic components assembly, as follows:

Uncompensated Turret Position	Voltage, Peak to Peak (vac, 400 Hz)	Compensated Turret Position for Weapon Selected	
		7.62mm	40mm
$30^{\circ} \pm 0.5^{\circ}$ left	1.5 ± 0.15	28 ± 4.6 MILS left of uncompensated turret position	98 ± 17 MILS left of uncompensated turret position
$30^{\circ} \pm 0.5^{\circ}$ right	1.5 ± 0.15	28 ± 4.6 MILS right of uncompensated turret position	98 ± 17 MILS right of uncompensated turret position

3.3.3.7 Override control. Actuation of the pilot override switch shall override the pilot controls including the Pilot/Gunner Select, Weapon Select, Wing Stores Select, and Master Arm switches. When actuated, the gunner shall be capable of firing by means of the Gunner Cyclic Stick with the weapons in the stow position. The Pilot Cyclic Stick and the Gunner Sighting Unit shall be inoperative with the exception of the Weapon Select switch.

3.3.4 Subsystem accuracy. The subsystem accuracy, excluding errors caused by range estimation, shall be such that 68% of the rounds fired shall group within an 8 mil radius of the boresight point aim (Figure 1) at 1,000 inches with either the 7.62mm or 40mm weapon. The subsystem accuracy shall not suffer degradation of more than 10% within a power range of 22 to 24 vdc. The subsystem shall be fired from a test stand which simulates the assembly interface between the M28A1 subsystem and the AH-1G helicopter as defined by Figure 2. The structural design of the test stand shall not degrade subsystem accuracy.

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3.3.5 Non-coincidence. The subsystem shall be incapable of firing when the angular difference between the sighting and the computed gun-line, excluding lead angle compensation, is $5^{o} + 3^{o} - 0^{o}$.

3.3.6 Compensation. Compensation shall not effect synchronization between the sight and weapons when inoperative.

3.3.6.1 Range compensation. The gun line deviation shall conform to the following values under the condition stated:

Gunner's compensation switch set at the IN position, pilot's sight range adjustment dial set at zero meters.

Gunner Range Adjustment Position (Meters)	Weapon	
	7.62mm (mils)	40mm (mils)
0	No greater than 3 mils from point at 0 meters with switch set at gunner's position with compensation out.	No greater than 3 mils from point at 0 meters with switch set at gunner's position with compensation out.
400	4 ± 3	21 ± 10
800	12.5 ± 8	64 ± 20

Gunner's compensation switch at the OUT or IN position - pilot in control switch activated.

Pilot Range Adjustment Position (Meters)	Weapon	
	7.62mm (mils)	40mm (mils)
0	No greater than 3 mils from point at 0 meters with switch set at gunner's position with compensation out.	No greater than 3 mils at 0 meters with switch set at gunner's position with compensation out.
400	4 ± 4	21 ± 14
800	12.5 ± 11	64 ± 30

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3.3.7 Reticle flasher. A reticle flasher circuit shall produce flashing of the reticle illumination lamps whenever noncoincidence between the sight line and gun line occurs and whenever the weapon is driven into the limit stop positions. The flasher circuit shall be activated at a weapon depression angle of 48° to 50°.

3.3.8 Turret position control.

3.3.8.1 Stow. The stow position is defined as the boresight position established in paragraph 3.3.13 and shall be repeatable within 6 mils. The weapons shall achieve a stow position upon actuation of the Master Arm Switch into the SAFE position, upon actuation of the Pilot Override Switch, or release of the Action Switch on the Sighting Unit. The stow position shall be electrically adjustable a minimum of five degrees, in azimuth and elevation, from the boresighted position.

3.3.8.2 Flexible control. The turret shall attain any position within the maximum azimuth and elevation envelope in response to commands from the gunner's sighting station controller.

3.3.8.3 Elevation counterbalance. The elevation counterbalance assembly shall return the weapons to the boresighted elevation position of 3.3.13, or greater, at any azimuth position in the event of hydraulic failure.

3.3.9 Field of fire.

3.3.9.1 Azimuth rotation angle. The turret shall attain any position to allow aiming and firing the installed weapons within a minimum angle of 105° left and right azimuth as measured from the stow position. Beyond 105° left and right azimuth position, the firing circuit shall be interrupted and the mechanical limit stops subsequently contacted. The mechanical limit stops shall limit the turret excursion to 110° maximum left and right azimuth. The turret shall meet all functional requirements within the maximum electrical excursion limits specified.

3.3.9.2 Elevation axis rotation angle. The turret shall attain any position to allow aiming and firing the installed weapons in a plane perpendicular to the azimuth plane within the limits of travel specified below:

a. Elevation limits.

(1) Electrical. The firing circuit shall be interrupted at 17.5° ± 1° between 57° ± 1° left and right azimuth and shall decrease linearly to a limit of 10.5° maximum at 110° left and right azimuth.

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(2) Mechanical. The elevation torque tube mechanism shall restrict maximum elevation to 20.5° ± 1° left and right azimuth, and shall decrease linearly to 12.7° maximum at 110° maximum left and right azimuth.

b. Depression limits. The depression limits shall be 50° minimum between 110° left and right azimuth.

3.3.10 Slew rates. The turret, powered as specified in 3.3.1 and 3.3.2, shall meet the following slew rate requirements:

a. Maximum azimuth slew time from limit to limit, in either direction, shall not exceed 3.066 seconds. Minimum azimuth angular (steady state) velocity shall not be less than 72° per second.
b. Maximum elevation and depression slew time from limit to limit, in either direction, shall not exceed 1.36 seconds. Minimum angular (steady state) velocity shall not be less than 60° per second.
c. Azimuth and elevation/depression slew time shall not degrade more than 10° with a decreased hydraulic input pressure of 1350 ± 25 psig.

3.3.10.1 Elevation smoothness. With the elevation command signal changing at a constant rate corresponding to a gun movement of 10° per second in elevation, the gun shall not oscillate or experience sudden incremental changes exceeding 2 milliradians.

3.3.10.2 Elevation velocity error. With the elevation command signal changing at a constant rate corresponding to a gun movement of 10° per second in elevation, the positional error shall not be more than 10 milliradians.

3.3.10.3 Azimuth smoothness. With the azimuth command signal changing at a constant rate corresponding to a turret movement of 10° per second in azimuth, the turret shall not oscillate or experience sudden incremental changes exceeding 2 milliradians.

3.3.10.4 Azimuth velocity error. With the azimuth command signal changing at a constant rate corresponding to a turret movement of 10° per second in azimuth, the positional error shall not be more than 10 milliradians.

3.3.11 Area fire. The turret shall oscillate in azimuth about the weapon aim point when operated in the Area Fire (Dither) mode.

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3.3.12 Limit switches.

3.3.12.1 Azimuth limit switching. Azimuth limit switches shall be activated at a minimum rotation angle of 105° left and right azimuth, which shall, upon actuation, interrupt weapon operation, and shall initiate reticle lamp flashing or pilot control indicator lamp flashing.

3.3.12.2 Elevation limit switch. The elevation limit switch shall interrupt weapon operation and produce reticle flashing.

3.3.12.3 Depressing limit switch. The depression limit switch shall initiate reticle flashing.

3.3.13 Boresighting. The gun line shall be boresighted to within 1 mil of the boresight point established on the boresight target (see Figure 1). After the turret/guns have been boresighted they shall be capable of being slewed both in azimuth and elevation and return to within a 6 mil radius of the recorded boresight starting points.

3.3.14 Weapons fire out. Each weapon shall cease operation whenever the corresponding last round of ammunition passes through the appropriate last round switch. The ammunition fire-out switch shall permit complete expenditure of ammunition.

3.3.15 Weapon clearing. With the clear/unclear switch in the clear position, the subsystem shall automatically clear a minimum of 7 to a maximum of 22 live rounds from the machine gun chamber and feed mechanism. The clear/unclear switch in the unclear position shall disable the clearing circuit when the conservation of ammunition is necessary.

3.3.16 Steady state rate of fire. The subsystem shall control the steady state rate of fire for each weapon as follows:

a. The 7.62mm machine gun shall operate selectively at a fixed steady state rate of fire of 4,000 ± 600, and 2,000 ± 300 rounds per minute, provided the firing voltage supplied at the input to the Weapon Speed Control unit is + 25 vdc ± 0.5 vdc, and provided the torque of the weapon is between 13 and 25 inch-pounds measured at the drive input. The assembly shall not be significantly degraded in performance nor suffer consequent damage when operated at a steady rate of fire for periods up to 6.0 seconds. The above requirements are predicated upon use of Government Standard 7.62mm, M80 ball cartridges or M62 tracer rounds.

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b. The 40mm, M129 grenade launcher shall operate at a steady state rate of fire of 400 + 40 - 60 rounds per minute, provided the firing voltage is as stated above and the 40mm weapon torque input is within the limits specified for the grenade launcher assembly. The M129 shall not be significantly degraded in performance nor suffer consequent damage when operated at a steady rate of fire for periods up to 11 seconds. This requirement is predicated upon use of Government Standard 40mm Practice M385 and High Explosive M384 ammunition utilizing standard M16 series links.

3.3.17 Time to firing. The time required to attain the specified steady state rate of fire for both the 40mm grenade launcher and 7.62mm machine gun shall not exceed 0.5 seconds.

3.3.18 Burst limiting. The firing circuit for the machine gun shall limit each burst to 6 ± 0.6 seconds. The firing circuit for the grenade launcher shall limit each burst to 10 ± 1.0 seconds. The weapons shall be operable immediately upon release and reactuation of the trigger switches.

3.3.19 40mm braking. The barrel of the 40mm grenade launcher shall stop in a safe position upon cessation of firing from any steady state rate of fire up to 440 rounds per minute with a maximum voltage input not to exceed 30 vdc. Safe stopping position is defined as non-intersection of the vertical plane established by the chamber end of the barrel with any part of the 40mm round. Upon release of the trigger, the dynamic brake shall enable the weapon to eject one dummy round, then fire one round, and stop in the required safe position.

3.3.20 Ammunition capacity. The capacity of the ammunition feed system shall consist of one 7.62mm ammunition rotary drum to provide a full complement of 4,000 ± 600 rounds to the 7.62mm machine gun, and one 40mm ammunition rotary drum to provide a full complement of 265 rounds to the M129 grenade launcher. An additional 18 rounds of 40mm ammunition may be loaded into the flexible ammunition chute making the total capacity of the feed system 283 rounds.

3.4 Performance characteristics.

3.4.1 Functioning. The subsystem shall be capable of operating without malfunction or failure when subjected to the functional requirements of 3.3.1 through 3.3.20. Function firing shall be accomplished using Government standard, 7.62mm, M80 ball cartridges and 40mm, M385 practice cartridges. Testing shall be as specified in 4.5.3.2.

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3.4.2 Endurance. The subsystem shall be capable of firing 28,000 rounds of Government standard, 7.62mm, M80 ball cartridges, and 1,981 rounds of Government standard 40mm, M385 practice cartridges for endurance with no more than one stoppage allowed for each weapon. A stoppage is defined as an event which prevents the subsystem from performing its intended function and cannot be immediately corrected by the pilot or gunner while in flight. Stoppages attributable to GFE shall be excluded unless caused by the subsystem. Testing shall be as specified in 4.7.

3.5 Interchangeability. Unless otherwise specified on the drawing, all parts shall be interchangeable. Testing shall be as specified in 4.5.3.3.

3.6 Aerial firing. When specified in the contract (see 6.2), the subsystem shall support the firing of 20,000 rounds of Government standard, 7.62mm, M80 ball cartridges, and 1,415 rounds of Government standard, 40mm, M385 practice cartridges with no more than one stoppage allowed for each weapon. A stoppage is defined as an event which prevents the subsystem from performing its intended function and cannot be immediately corrected by the pilot or gunner while in flight. Stoppages attributable to GFE shall be excluded unless caused by the subsystem. Testing shall be as specified in 4.4 and 4.6.3.

3.7 Environmental conditions. When specified in the contract (see 6.2), the subsystem shall perform as specified in 3.4.1 when subjected to the environmental conditions of 3.7.1 thru 3.7.7. There shall be no significant damage or deterioration of parts during or after subjection to the environmental conditions which might impair performance or service life. Testing shall be conducted in a manner which closely simulates the actual service use of the item being tested. Testing shall be as specified in 4.4 and 4.6.4.

3.7.1 High temperature. The subsystem shall operate in an environment equivalent to that specified in Method 501, Procedure I, of MIL-STD-810, with the following exceptions:

Maximum operating temperature: +160°F
Maximum non-operating temperature: +185°F

3.7.2 Low temperature. The subsystem shall operate in an environment equivalent to that specified in Method 502, Procedure I, of MIL-STD-810, at the following temperatures:

Operating temperature: -25°F
Non-operating temperature: -40°F
Time at non-operating temperature: 16 hours

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3.7.3 Vibration. The subsystem shall operate in an environment equivalent to that specified in Method 514.1, Category C, Procedure I, Curve M, of MIL-STD-810.

3.7.4 Shock. The subsystem shall not be adversely affected by an environment equivalent to a sawtooth shock pulse applied along the mutually perpendicular axes as defined in Method 516.1, Procedure I, of MIL-STD-810, with the following exceptions.

Axis	Magnitude	Duration
Vertical	4.5g	11 ± 1 MS (millisecond)
Major Horizontal	9.0g	11 ± 1 MS
Minor Horizontal	1.5g	11 ± 1 MS

3.7.5 Humidity. The subsystem shall withstand exposure to an environment equivalent to that specified in Method 507, Procedure I, of MIL-STD-810, except that the test duration shall be 168 hours.

3.7.6 Salt fog. The subsystem shall be capable of withstanding exposure to a salt atmosphere, equivalent to Method 509 of MIL-STD-810, without significant corrosion, clogging or binding of moving parts. This requirement shall not apply to the glass elements of the optical sight.

3.7.7 Dust. The subsystem shall not be adversely affected by exposure to airborne sand and dust equivalent to Method 510 of MIL-STD-810, with the following exceptions:

a. Air flow: 300 ± 50 feet per minute
b. Temperature: 77°F
c. Duration: 8 hours

This requirement shall not apply to the glass elements of the optical sight.

3.8 Reliability. When specified in the contract (see 6.2), the subsystem, excluding Government-furnished weapons and ammunition, shall have a minimum acceptable Mean-Mission-Between-Failure (MMBF) of not less than 2.00 missions, and a desired and specified MMBF of 3.00 missions. A mission is defined as the expenditure of 4,000 rounds of 7.62mm ammunition and 283 rounds of 40mm ammunition during a two-hour flight. A failure shall be defined as an event which prevents the subsystem from performing its intended function and cannot be immediately corrected by the pilot or gunner while in flight. Testing shall be as specified in 4.4 and 4.6.6.

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3.8.1 Reliability criteria. The above reliability requirement is based upon a minimum acceptable Mean-Round-To-Stoppage (MRTS) of 12,000 rounds for the 7.62mm portion and 2,000 rounds for the 40mm portion, with a minimum acceptable Mean-Time-Between-Failure (MTBF) of 69 hours for time sensitive components independent of the aforementioned MRTS requirements.

3.9 Electromagnetic interference (EMI). When specified in the contract (see 6.2), the subsystem shall conform to the requirements for class ID equipment of MIL-STD-461. Testing shall be as specified in 4.4 and 4.6.5.

3.10 Marking. The subsystem shall be marked in accordance with MIL-W-13855. The subsystem shall be identified by a serial number assigned by the procuring activity (see 6.2).

3.11 Workmanship. The workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality Assurance terms and definitions used herein are in accordance with MIL-STD-109 and MIL-STD-721.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.2). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein, and

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when specified in the contract, the aerial firing test (see 3.6 and 4.6.3), environmental test (see 3.7 and 4.6.4), reliability test (see 3.8 and 4.6.6), electromagnetic interference (EMI) test (see 3.9 and 4.6.5), and such other inspections as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Inspection lots of parts. The formation, size and presentation of inspection lots of parts shall be in accordance with MIL-STD-105. Inspection lots shall be as large as practicable, in consideration of quality history, manufacturing conditions, and the contractor's delivery schedule, and within limitations of MIL-W-13855.

4.5.2 Examination.

4.5.2.1 Component parts and concurrent repair parts. Examination of parts shall be performed in accordance with the criteria specified in the contract (see 6.2). The contractor's examination of parts shall be accomplished prior to their assembly in the end items or submission for acceptance as repair parts.

4.5.2.2 Subsystems. Visually and manually examine each subsystem to determine conformance with the material, construction and design requirements of 3.2. Each step in the examination shall include a visual examination for proper cleaning and presence of the specified protective coating and to determine the general quality, completeness of manufacture, assembly, clarity and legibility of markings and workmanship. The examination provisions shall be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Reinspections of these characteristics on the completed product is not required provided assurance exists that the characteristic has not been changed, degraded or damaged by subsequent manufacturing, assembly or handling and that adequate inspection records are maintained. Subsystems failing to meet the requirements shall be rejected.

4.5.2.2.1 Weight. One completely assembled subsystem shall be weighed to determine conformance with the requirements of 3.2.7.

4.5.2.2.2 Visually examine all markings for the subsystem and components for compliance with the applicable drawings, for secure assembly, serialization, and to assure that all pertinent data has been applied (see 3.10).

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4.5.3 Testing.

4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete subsystem. If test requirements specified herein are not met, acceptance of the subsystems shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine subsystems, partially assembled subsystems, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned subsystems for retest.

4.5.3.2 Functional testing. Each subsystem shall be tested in accordance with the test method specified in 4.6.1 to determine compliance with functional requirements of 3.4.1. Subsystems failing to meet the functional requirements shall be rejected.

4.5.3.3 Interchangeability testing. A sample of three subsystems, selected by the Government representative from each of the first three months production, and thereafter from each alternate month's production, shall be tested for interchangeability of parts (see 3.5) using the test method specified in 4.6.2. Subsystems selected for interchangeability shall have been found satisfactory in all other examinations and tests. The subsystem, after interchange of parts, shall be function tested using the test methods specified in 4.6.1. Failure of the subsystem to meet the interchangeability test shall be cause for rejection of the represented lot. Prior to submission as a reconditioned lot, the contractor's correction shall be effected on all subsystems of the lot and on all subsystems and component parts not yet accepted. Sample size and test method for a reconditioned lot shall be the same as for the original lot. Failure of the interchangeability test shall also be cause for the contractor to perform an interchangeability test on the next two month's production.

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4.5.3.4 Component parts and concurrent repair parts testing. Raw material testing, part testing, and certification shall be performed in accordance with the criteria specified in the contract (see 6.2). This shall include chemical analysis andphysical tests of materials, tests of protective finish, heat treatment, bonding and function of parts as applicable. The contractor shall accomplish these tests prior to assembly of parts into the end item.

4.5.4 Packaging examination and testing. Unless otherwise specified (see 6.2), the packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.5 Inspection equipment.

4.5.5.1 Acquisition, calibration, maintenance and disposition. Unless otherwise specified (see 6.2) responsibility for acquisition, calibration, maintenance and disposition of acceptance inspection and test equipment prescribed on the Quality Engineering Data List QE-G-DL11690269 and for all other inspection equipment required to perform inspection required by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.5.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetetive measurements to an accuracy of 10 percent of the total tolerance of the characterist cs being inspected.

4.6 Test methods.

4.6.1 Functional test. The subsystem shall be functionally tested in accordance with the Functional Test Procedure specified on QE-D-GL 11690269.

4.6.1.1 The functional test includes a live firing procedure for each of the four weapons configurations (dual 7.62mm, dual 40mm or mixed 7.62mm and 40mm). The functional test live firing procedure shall be as specified in the contract (see 6.2).

4.6.1.2 When the 7.62mm rotary ammunition drum is procured under this specification, the live firing portion of its functional test shall be accomplished during the ALL ANGLE FIRING portion of the subsystem tests. The 7.62mm drum shall be loaded in accordance with the functioning test of MIL-D-45929.

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4.6.2 Interchangeability test. Subsystems shall be tested for interchange of parts by disassembling and reassembling the parts, using the parts and the prearranged system specified below. Interchange of parts shall be accomplished by dividing the parts of each subsystem into three groups of nonmating parts as shown below and distributing the groups into three separate areas until each area contains parts for a complete subsystem. Groups of parts from the first subsystem shall be taken in order and placed in areas A, B and C respectively; groups of parts from the second subsystem shall be taken in order and placed in areas B, C and A respectively; groups of parts from the third subsystem shall be taken in order and placed in areas C, A and B respectively. Commercial parts such as screws, nuts, washers and pins shall be placed in the same area as their mating or associated part. Any commercial part rendered unserviceable by disassembly shall be replaced without penalty to the interchangeability test. Laminated shims may be replaced without penalty to the interchangeability test. The subsystems shall be reassembled using only those parts which are in the same area of nonmating parts. After interchange of parts, subsystems shall be function tested in accordance with 4.6.1.

GROUPS OF NONMATING PARTS

Group 1	Group 2	Group 3
11687079	11687084	11687085
* 11687106	* 11687104	** 11687098
11687108	11687107	* 11687105
11687122	11687112	** 11687110
11687123	11687116	** 11687111
11681137	11687128	11687114
11687180	11687163	11687118
11687193	11687210	11687136
11687199	11687212	11687140
11687224	11687214	11687203
11687230	11687223	11687216
11687233	11687225	11687217
11687241	11687267	11687219
11687252	11687342	11687226
11687254	11687384	11687229
11687255	11687386	11687258
11687256	11687420	11687380
11687257	11687461	11687383
11687266	11690251	11687387
11687374	11690264	11687389
11687381	11691215	11687393

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GROUPS OF NONMATING PARTS (cont)

Group 1	Group 2	Group 3
11687385		11687403
11687392		11687413
11687874	Subsystem less parts	11687421
11688532	specified in Groups 1, 2	11687457
11690071	and 3 hereon.	11687467
11690252		11687987
11690253		11688148
11691456		11688531
		11689556
		** 11690145
		11690263
		11690265
		11690300
		11691270

** Parts noted may remain together with the gearshaft (11687108) because the bearing (11010124-2) cannot be removed without sustaining damage.

* Bearings 11014596-12 may remain with noted parts to prevent damage to the bearings which would result from disassembly.

NOTE: Part numbers 11687190, 11687191 and 11690267 may remain with the saddle to which they are attached.

4.6.3 Aerial firing test. When specified in the contract (see 6.2), the subsystem weapon configuration for each aerial firing test cycle shall be either R/H 7.62mm - L/H 40mm or R/H 40mm - L/H 7.62mm. The weapon configuration may vary between the test cycles and shall be as specified in the contract (see 6.2). The subsystem shall be mounted in a helicopter and be tested five cycles aerially in accordance with Table I to determine compliance with the requirement of 3.6. Each cycle shall consist of firing a full complement of 7.62mm and 40mm ammunition as specified in 3.3.20, in accordance with the schedule listed in Table I. Each complement shall be initiated at position I and continued through the positions until the complement is exhausted. When the last-round switch interrupts firing, position the ammunition fire-out switch on the gunner control panel to ON and continue. Due to the weapon rate-of-fire variances, the complement may be expended prior to completing all positions. When all positions have been fired and part of the complement remains, refire the Table I schedule, starting with position 1 and continue until the ammunition is exhausted.

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4.6.3.1 In the event that a burst is fired at the wrong rate, mode or duration, the sequence shall be continued. When a condition is met during the firing cycle that coincides with the prior erroneous burst, it (the burst(s)) shall not be repeated at that time, and the omitted bursts shall be fired at the end of the sequence.

4.6.3.2 Disposition instructions for aerial fired tested subsystems shall be as specified in the contract (see 6.2).

Symbols used in Table I for the aerial firing schedule are defined as follows:

Aircraft

VNE - Velocity
MRC - Maximum rate of climb
MRD - Maximum rate of descent
MA - Maximum acceleration
MDe - Maximum deceleration
CR - Cruising range 80 - 100 knots
H - Hover

Weapon Position

MLA - Maximum left azimuth (before electrical shutoff)
MRA - Maximum right azimuth (before electrical shutoff)
ME - Maximum elevation (before electrical shutoff)
MD - Maximum depression
STOW- (see 3.3.8.1)

Pilot/Gunner

P - Pilot in control
G - Gunner in control

Weapon mode

U - Unclear
C - Clear

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Table I - Aerial Firing Schedule

	WEAPON AZIMUTH	WEAPON ELE/DEP	WEAPON CALIBER	APPROX. NO. OF ROUNDS	SECONDS OF FIRE	FIRING RATE	WEAPON MODE	PILOT OVERRIDE SWITCH	PILOT/ GUNNER	ARCFT
1.	0°	STOW	7.62mm	200	Burst Limit	Low	U	Off	P	H
2.	0°	STOW	7.62mm	200	Burst Limit	Low	U	On	P	VNE
3.	MLA to MRA	ME to MD	7.62mm	200	Burst Limit*	Low	C	Off	G	MRC
4.	MLA to MRA	0°	40mm	24	4			Off	G	MA
5.	MRA to 0°	0° to MD	40mm	30	5**			Off	G	MRD
6.	MLR to MRA	STOW to ME	7.62mm	200	3	High	U	Off	G	VNE
7.	MRA to MLA	ME to MD	7.62mm	200	3	High	U	Off	G	VNE
8.	MLA	MD	7.62mm	100	3	Low	U	Off	G	MDe
9.	MRA	MD	7.62mm	100	3	Low	U	Off	G	CR
10.	0°	STOW	7.62mm	200	3*	High	U	Off	P	MA
11.	0°	STOW	7.62mm	200	3	High	C	On	P	MA
12.	MRA to MLA	MD to ME	40mm	30	5			Off	G	CR

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Table I - Aerial Firing Schedule (Cont.)

	WEAPON AZIMUTH	WEAPON ELE/DEP	WEAPON CALIBER	APPROX. NO. OF ROUNDS	SECONDS OF FIRE	FIRING RATE	WEAPON MODE	PILOT OVERRIDE SWITCH	PILOT/ GUNNER	ARCRFT
13.	MLA to MRA	ME to STOW	40mm	24	4				G	MRD
14.	0°	STOW	40mm	60	Burst Limit			Off	P	MA
15.	MRA	ME	7.62mm	200	3*	High	U	Off	G	MRC
16.	MRA	ME	7.62mm	100	3*	Low	U	Off	G	CR
17.	MRA to MLA	ME to MD	7.62mm	400	Burst Limit	High	C	Off	G	MA
18.	MLA to MRA	MD to ME	40mm	60	Burst Limit			Off	C	MDe
19.	MRA	ME	7.62mm	200	Burst Limit	Low	U	Off	G	MRC
20.	MRA to MLA	ME to STOW	7.62mm	200	Burst Limit	Low	U	Off	G	CR
21.	0°	STOW	40mm	48	8			Off	P	MA
22.	0°	STOW	40mm	remainder				On	P	H
23.	MLA	MD	7.62mm	400	Burst Limit*	High	C	Off	G	MRD
24.	MLA	ME	7.62mm	200	Burst Limit*	Low	U	Off	G	H
25.	MLA to MRA	Me	7.62mm	200	Burst Limit	Low	U	Off	G	MRC

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Table I - Aerial Firing Schedule (Cont.)

	WEAPON AZIMUTH	WEAPON ELE/DEP	WEAPON CALIBER	APPROX. NO. OF ROUNDS	SECONDS OF FIRE	FIRING RATE	WEAPON MODE	PILOT OVERRIDE SWITCH	PILOT/ GUNNER	ARCRFT
26.	MRA	ME to MD	7.62mm	400	Burst Limit*	High	U	Off	G	MDC
27.	MRA to 0°	MD to STOW	7.62mm	100	3	Low	U	Off	G	MA

NOTES:

*Velocity Compensation IN.
**Range Compensation to 1600 meters and compensation IN.

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4.6.4 Environmental test. When specified in the contract (see 6.2), using the test equipment specified on QE-G-DL11690269 and a mixed (7.62mm and 40mm) weapon configuration, the subsystem shall be tested as specified in Table II (see 3.7).

4.6.4.1 Operational test.

a. Position the master arm switch on the pilot control panel to safe, and the gunner pilot control switch to PILOT. Observe that the pilot and gunner control panels' SAFE indicators and the PILOT IN CONTROL indicator on the sighting station illuminate.
b. Place the master arm switch on the pilot control panel to ARMED. Observe illumination of ARM indicators and appropriate 40mm weapon and 7.62mm weapon indicators on the pilot and gunner control panels.
c. Position the fire switch on the pilot control panel to POINT and the weapon select switch to LEFT. Actuate a switch simulating the pilot cyclic stick trigger. The left weapon shall operate. Release the trigger.
d. Repeat paragraph 4.6.4.1c with the pilot weapon select switch to RIGHT. The right weapon shall operate.
e. Repeat paragraph 4.6.4.1c, except place the weapon select switch to BOTH. Only the 7.62mm weapon shall operate.
f. Position the fire switch on the pilot control panel to AREA and actuate the simulated pilot cyclic stick trigger. The turret shall oscillate in azimuth and the 7.62mm weapon shall operate. Release the trigger. Place the pilot control panel fire switch to POINT.
g. Position the gunner pilot control switch on the pilot control panel to GUNNER, the fire switch on the gunner control panel to AREA, and depress the action switch; the subsystem shall oscillate in azimuth. Release the action switch and position the gunner control panel fire switch to POINT.
h. Actuate action switch and right firing trigger while moving sighting station in azimuth and elevation. The turret shall follow the movement of the sighting station and the 7.62mm weapon shall operate. Release the right firing trigger and repeat weapon operation using the left firing trigger; the 7.62mm weapon shall operate.
i. With the sighting station weapon select switch, select the 40mm weapon and depress the left 40mm firing button; the 40mm weapon shall operate. Release the left firing button and repeat weapon operation using the right firing button. The 40mm weapon shall operate. Position the weapon select switch to BOTH.

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j. Pin sighting in stow position; place the sight compensation switch to IN and the sight range adjust control to 1600 METERS. The weapon shall elevate from a stowed position. Release action switch and unpin sight. Place compensation to OUT and range adjust control to 0 METERS.

k. Position the filament select switch on the sighting station to each position and observe that the reticle lamp is on for both positions.

l. Rotate the reticle intensity control from OFF to full clockwise position; the reticle shall vary in intensity from OFF to full brightness.

m. Place the pilot override on the gunner control panel to ON and the pilot control panel master arm switch to OFF. Depress the action and trigger switches on the sighting station. The 7.62mm weapon shall not operate and the turret shall remain in a stow position. Release the action and trigger switches. With the pilot weapon select switch in left and the sighting station weapon select switch in right, actuate the simulated pilot cyclic stick trigger. The weapon shall not operate. Release the pilot cyclic stick trigger and actuate the sighting action switch and firing switches and buttons. The weapons shall not operate. Release the sighting station action and firing switches. Actuate the simulated gunner cyclic stick trigger. The right weapon shall operate while remaining in the stow position. Position the pilot override switch to OFF. The subsystem performance shall meet the requirements specified in 3.3.3.7.

n. Place the pilot control panel master arm switch to SAFE and depress the sighting station action switch. Rotate the sighting station from left azimuth to right azimuth. Repeat from right to left. The velocity and slewtime shall meet the requirements specified in 3.3.10.

o. Rotate the sighting station from elevation to depression. Repeat from depression to elevation. The velocity and slewtime shall meet the requirements specified in 3.3.10.

p. With an elevation command signal changing at a constant rate of 10° per second in elevation, the turret shall meet the smoothness requirements specified in 3.3.10.1. The error shall not exceed that specified in 3.3.10.2.

q. With an azimth command signal changing at a constant rate of 10° per second in azimuth, the turret shall meet the smoothness requirements specified in 3.3.10.3. The error shall not exceed that specified in 3.3.10.4.

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TABLE II - ENVIRONMENTAL TESTING

ENVIRONMENT	TEST SEQUENCE	OPERATIONAL TEST SCHEDULE (SEE 4.6.4.1)	FIRING TEST SCHEDULE (SEE 4.6.4.2)	TEST METHOD/PROCEDURE & REQUIREMENT PARA. NO.
Low Temperature	First	Before-During-After	Before-After	3.7.2
High Temperature		Before-During-After		3.7.1
Dust	Second	Before*-After	Before*-After	3.7.7
Shock	Third	Before*-During-After	Before*-After	3.7.4
Vibration	Fourth	Before*-During-After	Before*-After	3.7.3
Humidity	Fifth	Before*-After	Before*-After	3.7.5
Salt-Fog	Sixth	Before*-After	Before*-After	3.7.6

* At the option of Contracting Officer, these tests may be waived.

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4.6.4.2 Ground firing test. Load the subsystem with a full complement of 7.62 and 40mm ammunition. Apply electrical and hydraulic power as specified in 3.3.1 and 3.3.2. Position the pilot control panel master arm switch to ARM. Use the test set low and high fire trigger switches to fire from the stow positions and the sighting station triggers at all other positions. When firing from the sighting station, fire while slewing the weapons. Example: For position 2, the burst is fired while slewing from 105°R azimuth to 105°L azimuth, etc., except as otherwise noted. Each complement shall start with position 1 and continue until the complement is exhausted. Due to the variances in the weapon's rate of fire, the ammunition may be expended prior to completing all positions. When all positions are fired and part of the complement is remaining, refire the schedule starting with position 1 and continue until the ammunition is exhausted. When the last rounds switch interrupts firing, position the ammo fire-out switch on the gunner control panel to ON and continue the firing in accordance with Table III - Firing Schedule.

4.6.4.2.1 In the event that a burst is fired at the wrong rate, mode or duration, the sequence shall be continued. When a condition is met during the firing cycle that coincides with the prior erroneous burst, it (the burst(s)) shall not be repeated at that time, and the omitted bursts shall be fired at the end of the sequence.

4.6.4.3 Disposition of environmental tested subsystems shall be as specified in the contract (see 6.2).

4.6.5 Electromagnetic interference test. When specified in the contract (see 6.2), the subsystem shall demonstrate electromagnetic compatibility when operated in accordance with the functional procedures of 4.5.3.2 and tested in accordance with MIL-STD-462 to determine compliance with requirements of 3.9. Disposition of EMI tested subsystem shall be as specified in the contract (see 6.2).

4.6.6 Reliability test. When specified in the contract (see 6.2), the reliability tests demonstrating the requirements of 3.8 shall be conducted with a 7.62mm weapon and a 40mm weapon installed in the turret. For each test cycle, defined in 4.6.6.1, the failure versus test cycles data shall be plotted. One test cycle shall represent one mission. Testing shall be terminated whenever a decision point as defined in figure 3 for the subsystem has been reached. When firing during a ground reliability test, the subsystem (with sight) shall be mounted on a Government-approved firing fixture which simulates the aircraft installation.

4.6.6.1 Firing test procedure. Apply electrical and hydraulic power, as specified in 3.3.1 and 3.3.2, to the system and perform an operational test in accordance with 4.6.4.1 (excluding n through q). A test cycle (mission) shall consist of two hours of ON time for the system electronics, 283 rounds of 40mm and 4,000 rounds of 7.62mm ammunition fired while performing the test specified in 4.6.4.2. If the actual firing test time per cycle is under two hours, the system electronics must remain on

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TABLE III - FIRING SCHEDULE

POSITION NUMBER	PILOT OVERRIDE SWITCH	PILOT/ GUNNER SWITCH	WEAPON POSITION AZ		EL/DEP	WEAPON FIRING	APPROXIMATE NO. OF ROUNDS	SECONDS OF FIRE	RATE OF FIRE	WEAPON MODE
1	Off	Pilot	Stow			7.62mm	200	Burst Limit	Low	Unclear
							COOL			
2	Off	Gunner	105°R	to	-50°	7.62mm	200	Burst Limit	Low	Unclear
			105°L		-50°					
							COOL			
3	Off	Gunner	105°L	to	-50°	7.62mm	200	Burst Limit	Low	Clear
			105°R		+12°					
							COOL			
4	Off	Gunner	105°R	to	+12°	40mm	24	4.0		
			105°R		-50°					
5*	Off	Gunner	105°R		-50°	40mm	30	5.0		
							COOL			
6	Off	Gunner	105°R	to	-50°	7.62mm	200	3.0	High	Unclear
			105°L		+12°					
7*	Off	Gunner	105°L		+12°	7.62mm	200	3.0	High	Unclear
							COOL			

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TABLE III - FIRING SCHEDULE

POSITION NUMBER	PILOT OVERRIDE SWITCH	PILOT/ GUNNER SWITCH	WEAPON POSITION AZ	EL/DEP	WEAPON FIRING	APPROXIMATE NO. OF ROUNDS	SECONDS OF FIRE	RATE OF FIRE	WEAPON MODE
8	Off	Gunner	105°L to 105°R	+12° to -50°	7.62mm	100	3.0	Low	Unclear
9.	Off	Gunner	105°R	-50°	7.62mm	100	3.0	Low	Unclear
						COOL			
10	Off	Gunner	0°	0°	7.62mm	100	1.5	High	Unclear
11	On	Pilot	Stow		7.62mm	100	1.5	High	Unclear
						COOL			
12	Off	Pilot	Stow		7.62mm	100	3.0	Low	Clear
13	On	Pilot	Stow		7.62mm	100	3.0	Low	Clear
						COOL			
14	Off	Gunner	0°	0°	40mm	30	5.0		
						COOL			
15	Off	Gunner	105°L to 0°	+12° to 0°	40mm	24	4.0		
16	Off	Pilot	Stow		40mm	30	5.0		
17	On	Pilot	Stow		40mm	30	5.0		
						COOL			

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TABLE III - FIRING SCHEDULE

POSITION NUMBER	PILOT OVERRIDE SWITCH	PILOT/ GUNNER SWITCH	WEAPON POSITION AZ		WEAPON POSITION EL/DEP		WEAPON FIRING	APPROXIMATE NO. OF ROUNDS	SECONDS OF FIRE	RATE OF FIRE	WEAPON MODE
18*	Off	Gunner	0°		0°		7.62mm	200	3.0	High	Unclear
19*	Off	Gunner	105°		+12°		7.62mm	100	3.0	Low	Unclear
								COOL			
20	Off	Gunner	105°R	to	+12		7.62mm	400	Burst Limit	High	Clear
			105°L		-50°						
								COOL			
21	Off	Gunner	105°L	to	-50°		40mm	60	Burst Limit		
			105°R		+12°						
								COOL			
22	Off	Gunner	105°R	to	+12°		7.62mm	200	Burst Limit	Low	Unclear
			105°L		-50°						
								COOL			
23	Off	Gunner	105°L	to	-50°		7.62mm	200	Burst Limit	Low	Unclear
			105°L		+12°						
								COOL			
24	Off	Gunner	105°L	to	+12°		40mm	30	5.0		
			105°L		-50°						
25	Off	Gunner	105°L	to	-50°		40mm	Remainder			
			0°		0°						

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TABLE III - FIRING SCHEDULE

POSITION NUMBER	PILOT OVERRIDE SWITCH	PILOT/ GUNNER SWITCH	WEAPON POSITION AZ	WEAPON POSITION EL/DEP	WEAPON FIRING	APPROXIMATE NO. OF ROUNDS	SECONDS OF FIRE	RATE OF FIRE	WEAPON MODE
26	Off	Pilot	Stow		7.62mm	200	3.0	High	Clear
27	On	Pilot	Stow		7.62mm	200	3.0	High	Clear
						COOL			
28*	Off	Gunner	0°	0°	7.62mm	200	Burst Limit	Low	Unclear
						COOL			
29	Off	Gunner	105°L to 105°R	+12° to +12°	7.62mm	200	Burst Limit	Low	Unclear
						COOL			
30	Off	Gunner	105°R to 105°R	+12° to -50°	7.62mm	400	Burst Limit	High	Unclear
						COOL			
31	Off	Gunner	105°R to 105°R	-50° to +12°	7.62mm	100	3	Low	Unclear

Release action switches and position the fire-out switch to OFF.

* Firing shall be conducted with weapons held stationary at this position.

COOL Continuous forced air cooling during all firing tests is permissible at the option of the contractor. However, as a minimum, during all firing tests, a cooling period of not less than three minutes - to reduce the temperature of the gun - using forced air, shall be observed after each burst limit firing.

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whatever time is necessary to achieve an average of 2 hours total ON time per cycle. Failures occuring during this time (excluding those attributable to weapons and ammunition) shall be considered system failures and will, therefore, affect the outcome of the test.

4.6.6.1.1 At the option of the contractor, the test may be interrupted during or at the end of a cycle and the test restarted later at the point at which it was interrupted. The restart time shall not exceed 65 hours unless approved by the procuring agency.

4.6.6.1.2 Complete, accurate records shall be kept for each reliability test, showing each malfunction, stoppage and part replacement, including the number of the ammunition round at which each occurred. In addition, complete, accurate records shall be kept for all time losses attributed to the subsystem, including the time for each stoppage.

4.6.6.1.3 At the end of each test cycle, open subsystem fairing doors and visually examine the interior for any deterioration.

4.6.6.2 Disposition instructions for reliability tested subsystems shall be as specified in the contract (see 6.2).

4.7 Endurance test. Perform endurance testing as specified in the contract (see 6.2), using the procedures of 4.6.4.2, 4.6.4.2.1 and Table III - Firing Schedule, 7 consecutive times, to determine compliance with the requirements of 3.4.2.

4.7.1 Disposition instructions for endurance tested subsystems shall be as specified in the contract (see 6.2).

5. PACKAGING

5.1 Pilot pack. The pilot pack shall consist of a complete subsystem packaged in accordance with Packaging Data Sheet P11691612-1 for the level of protection specified in the contract (see 6.2), packed level B and forwarded in accordance with 3.1.

5.2 Levels A and B. The subsystem shall be preserved, packed and marked in accordance with Packaging Data Sheet P11691612-1 for the level of protection specified in the contract (see 6.2).

5.3 Commercial. Commercial preservation, packing and marking shall be in accordance with Fed. Std. No. 356.

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5.4 Repair parts. Repair parts shall be preserved, packed and marked in accordance with the Packaging Data Sheet or other requirements and for the level of protection specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The M28A1 Helicopter Armament Subsystem is intended for, but not limited to, use on the AG-1G Huey Cobra Helicopter. The armament subsystem shall mount and operate either a M134 machine gun, or a twin 40mm, M129 grenade launcher, or shall consist of one each; 7.62mm, M134 machine gun and 40mm, M129 grenade launcher, that may be installed in either right or left-hand position.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. List of drawings, specifications and publications pertinent to the subsystem showing applicable revision dates.
c. Shipping instructions for first article, and pilot pack (see 3.1, 4.4 and 5.1).
d. Aerial firing test, when required, and selection of applicable weapons configuration (see 3.6, 4.4 and 4.6.3).
e. Environmental condition test, when required (see 3.7, 4.4 and 4.6.4).
f. Reliability test, when required (see 3.8, 4.4 and 4.6.6).
g. Electromagnetic interference (EMI) test, when required (see 3.9, 4.4 and 4.6.5).
h. Block of serial numbers (see 3.10).
i. Criteria for examination and testing of component parts and concurrent repair parts (see 4.5.2.1 and 4.5.3.4).
j. Packaging examination and testing, if different (see 4.5.4).
k. Responsibilities for acquisition, maintenance and disposition of inspection equipment, if other than as specified (see 4.5.5.1).
l. Requirements for functional live firing procedure (see 4.6.1.1).
m. Disposition of subsystems used in aerial firing, environmental, electromagnetic interference (EMI), reliability and endurance testing (see 4.6.3.2, 4.6.4.3, 4.6.5, 4.6.6.2 and 4.7.1).
n. Requirements for, and the selection of, subsystems for endurance testing (see 4.7).
o. Selection of the applicable level of protection (A or B), or commercial packaging (see 5.1, 5.2 and 5.3).
p. Packaging instructions for repair parts (see 5.4).

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6.3 Supersession data. This specification includes the requirements of U.S. Army Weapons Command Purchase Description USAWECOM PD-26A dated 31 March 1972.

6.4 Depression limit stop. The mechanical depression limit stop should only be effected when the hydraulic activating cylinder is driven into the zero actuated position.

6.5 Useful life. The normal operating time to wearout of the subsystem should not be less than 1,000,000 rounds of 7.62mm or 400,000 rounds of 40mm or a proportional combination of both for each gun installation in the turret. This useful life figure is based upon the implementation of scheduled replacement of components including weapons when deemed necessary. The normal operating life of each subsystem is 1,200 flying hours in addition to the above rounds fired.

Custodian:

Army - WC

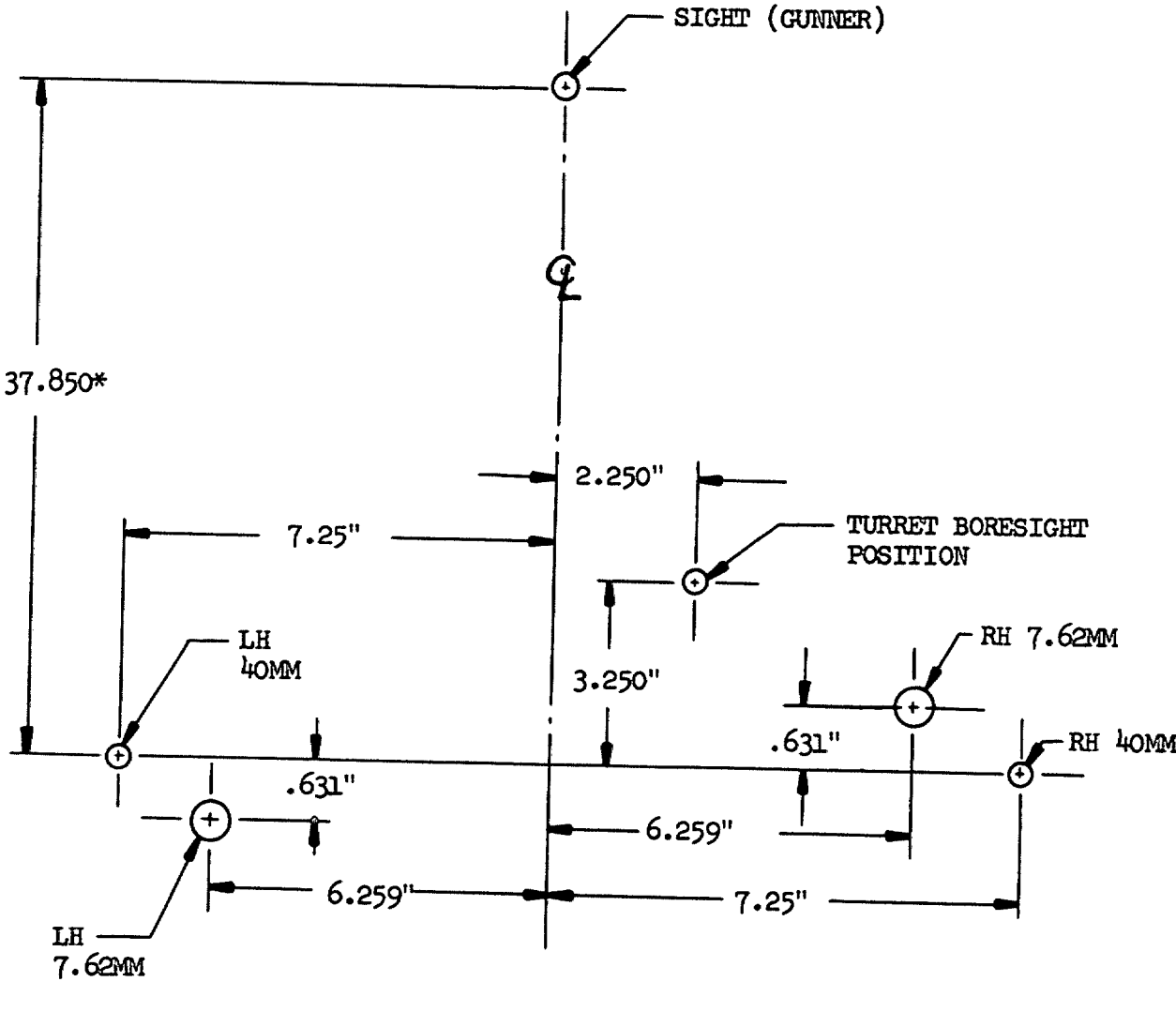
Preparing activity:

Army - WC

Project number:

1010-A602

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* This Dimension Depends Upon the Test Stand Utilized

BORESIGHT TARGET
at 1,000 inches

FIGURE 1

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NOTE: For fabrication of the firing test stand all station, waterline, buttline dimensions marked ⊖ and locating dimensions, other than datums A, B, C, & D for basic turret and weapon locations, apply with a ± .13 tolerance.

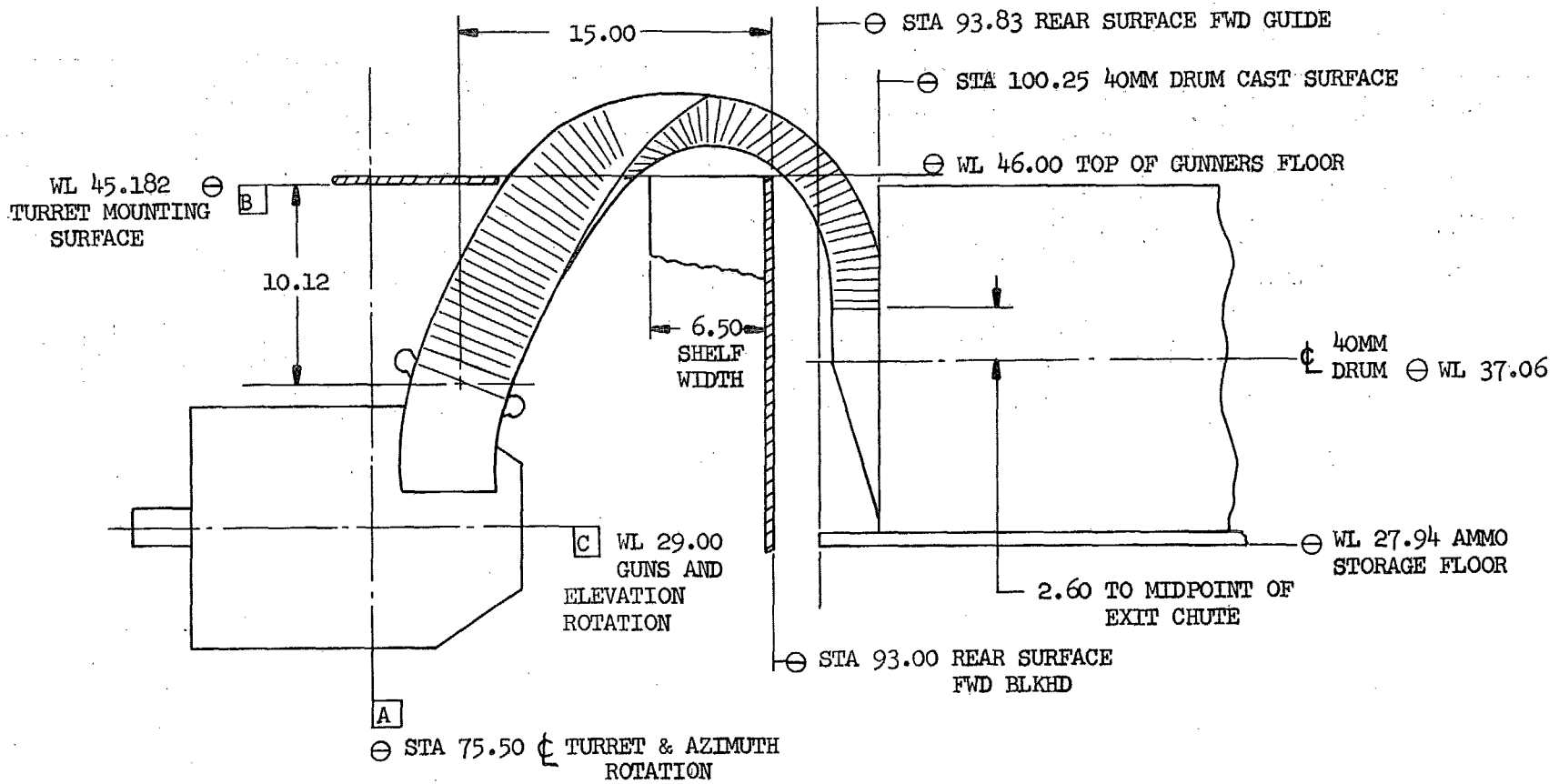


FIGURE 2 (SHEET 1 of 2)
RELATIONSHIP OF M28A1 SUBSYSTEM TURRET AND AMMO DRUMS ON AH-1G HELICOPTER
(40MM FLEX CHUTE CONF (0° AZ 0°EL POS) SHOWN)

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TOP VIEW SHOWING BUTTLINE LOCATIONS OF WEAPONS & AMMO DRUMS

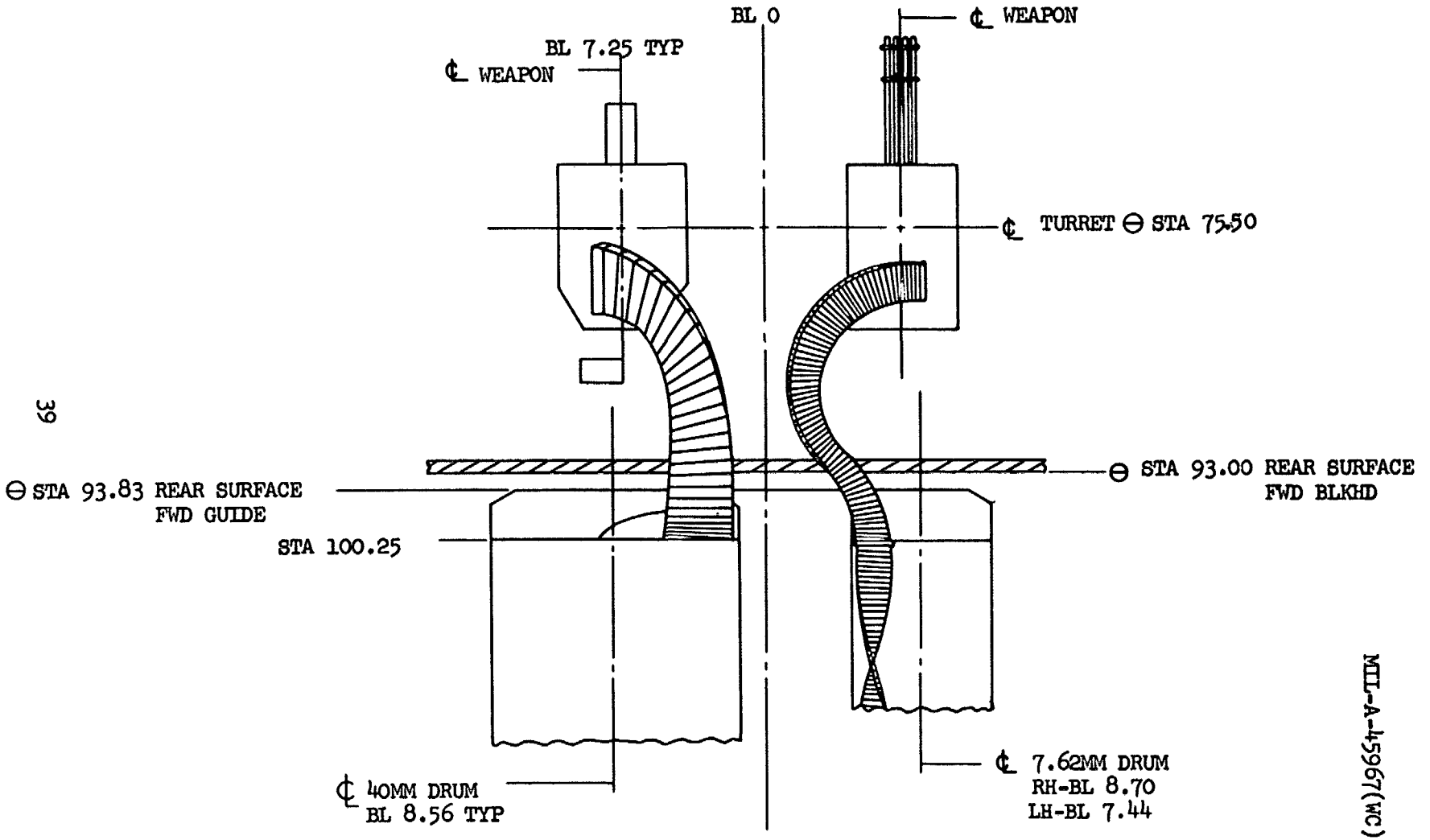
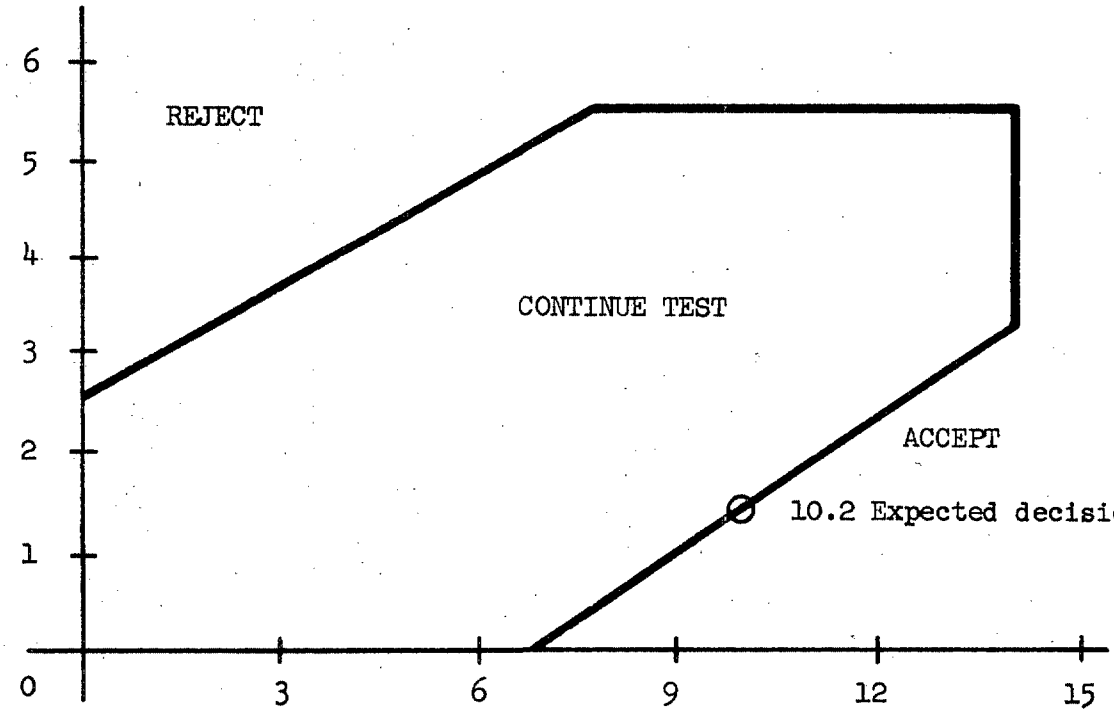


FIGURE 2 (SHEET 2 of 2)

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DECISION RISKS: 30%
DISCRIMINATION RATIO: 1.5:1
MIL-STD-781, TEST PLAN VII

RELIABILITY ACCEPTANCE TESTING



NO. OF FAILURES	REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	N/A	7
1	N/A	9
2	N/A	12
3	3	14
4	5	14
5	8	14
6	14	N/A

FIGURE 3

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-A-45967; Armament Subsystem, Helicopter: M28A1

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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